

Katarina Andrade

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EDUCATION

Northwestern University, Evanston, IL **Expected:** Jun 2027
Bachelor of Science in Communication
Majors in Theatre & Physics (Astronomy concentration) GPA: 3.86/4.0
Relevant coursework: Creating Inclusive Science; Observational Astrophysics; Extragalactic Astrophysics and Cosmology; Science Communication

RESEARCH EXPERIENCE

Active Galactic Nuclei Group, Yale University *Jun 2025 – Sep 2025*
Research Assistant
Advisor: Dr. Meg Urry
Analyzed JWST/NIRSpec spectra of high-redshift galaxies ($z > 4$) to identify AGN signatures using [Ne V] emission; built a pipeline for source selection based on color, compactness, and emission-line detectability.

Observational Galaxy Evolution Group, Northwestern University *Jun 2024 – Jun 2025*
Research Assistant
Advisor: Dr. Allison Strom
Improved spectral stacking methods to reveal faint emission lines in ~ 50 high-redshift ($z \approx 2 - 3$) galaxies; analyzed Keck/KBSS and JWST CECILIA (Chemical Evolution Constrained using Ionized Lines in Interstellar Aurorae) survey data across the H, J, and K bands.

Northwestern School of Communication's Molosky Summer Award 2026
Northwestern's Summer Internship Grant Program (SIGP) 2025
Alumnae of Northwestern University Undergraduate Research Scholar 2024
Northwestern Academic Year Undergraduate Research Grant (AYURG) 2024
Northwestern Summer Undergraduate Research Grant (SURG) 2024

PRESENTATIONS

Talk, Summer Research Symposium, "A Spectroscopic Analysis of the Little Red Dots" 2025
Yale Department of Astronomy, USA
Poster, Northwestern Undergraduate Research Expo, "Let Them Shine: Revealing Faint Galaxies with the Spectral Stacking Technique" 2025
Northwestern University, USA
Talk, Observational Galaxy Evolution Group Meeting, "Analyzing faint emission lines in distant galaxies through stacks" 2025
Northwestern University, USA

OUTREACH EXPERIENCE

REACH: Research Experiences in Astronomy at CIERA for High School Students *Mar 2026 – Sep 2026*
Instructional Support Mentor
Assisted students through coding training modules and educational activities, including telescope viewing at Dearborn Observatory; mentored a student through a one-on-one research project over three weeks at the

Center for Interdisciplinary Exploration and Research in Astrophysics (CIERA).

Wright Lab, Yale University

Jul 2025

Science Outreach Volunteer

Led a hands-on “Making Sound and Heat Visible,” activity for Pathways to Science Summer Scholars, summer program for high school students.

WiP (We’re in Physics), Northwestern University

May 2024 – Present

Co-President

Led Northwestern’s club supporting undergraduate women and gender minorities in physics; assisted students in attending conferences, engaging in research, and receiving mentorship.

Dearborn Observatory, Northwestern University

Nov 2023 – Present

Host

Hosted public observing sessions by operating 18.5" refracting telescope and coordinating Public Night tours.

Sciety Lab Institute, Brazil

Jan 2022 – Dec 2023

Co-founder and Coordinator

Chaired a remote nonprofit institute engaging middle school, high school, and undergraduate female-identifying students in STEM through the program Mulheres e Meninas nas Galáxias (Women and Girls in Galaxies); organized galaxy morphology classification projects, panels with astronomers, and scientific writing workshops.

TEACHING EXPERIENCE

Northwestern Center for Talent Development

Jan 2026 – Present

Teaching Assistant

Facilitated STEM learning for Pre-K and Grades 1–2 students in enrichment courses including Treasure Hunt: Math and Maps and Kids Chemists; Guided hands-on experiments, adapting scientific concepts into age-appropriate lessons and incorporating accommodations for students with learning disabilities.

Perez Elementary School

Jan 2024 – Apr 2024

Classroom Volunteer

Collaborated with lead Pre-K teachers to facilitate scientific, fine arts, and performing arts activities; delivered engaging lessons aligned with Perez’s arts-integrated curriculum.

EcossiSTEMa Jovem, Sergipe, Brazil

Feb 2022 – Dec 2023

Founder and Director

Headed the first official student club at school, aiming to present and discuss fun and day-to-day STEM topics for younger students; planned and taught lectures and classes to over 100 students.

SKILLS & INTERESTS

Technical Skills Python, PyQSOfit, GitHub, telescope instrumentation, science communication

Languages Portuguese (native), English (fluent), Spanish (advanced), French (beginner)

Interests Evolution of black holes and galaxies, Little Red Dots, Active Galactic Nuclei (AGN), Galaxy Evolution, Emission-line diagnostics, Women and FGLI inclusion in STEM, Scientific Outreach